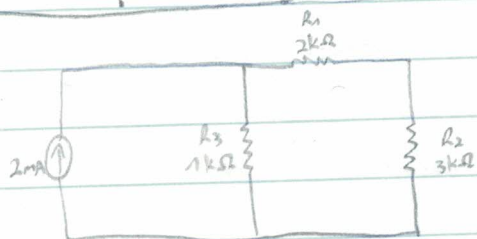
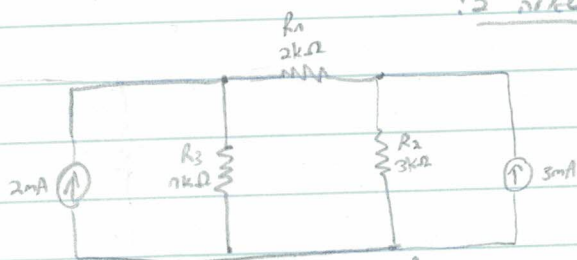


① 324394696 - 312 G.N

נתון: $I_{S1} = 2\text{mA}$

$I_{S2} = 3\text{mA}$

הנני מנסה



$R_{1,2} = 2 + 3 = 5\text{k}\Omega$; נכון $R_{1,2} = R_1 + R_2$; נכון $R_2 = 3\text{k}\Omega$; נכון $R_1 = 2\text{k}\Omega$
 $R_T = R_{1,2,3} = \frac{5 \cdot 1}{5 + 1} = \frac{5}{6}\text{k}\Omega$; נכון $R_{1,2,3} = \frac{R_{1,2} \cdot R_3}{R_{1,2} + R_3}$; נכון $R_3 = 1\text{k}\Omega$; נכון $R_{1,2} = 5\text{k}\Omega$

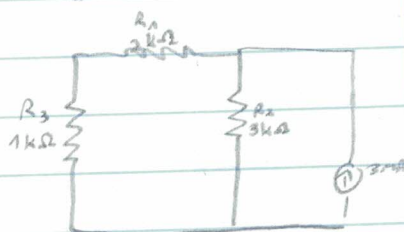
הנני מנסה להחליט על הזרם I_{R1} ו- I_{R2}

$I_{R1,2} = 2 \cdot \frac{1}{6} = \frac{2}{6}\text{A}$; $I_{R1,2}$

$\downarrow$
 $I_{R1} = I_{R2} = \frac{2}{6}\text{A}$

נכון $R_2 = 3\text{k}\Omega$; נכון $R_1 = 2\text{k}\Omega$

$\downarrow$
 $I_{R3} = 2 \cdot \frac{5}{6} = \frac{5}{3}\text{A}$; I_{R3}



$R_{1,3} = 1 + 2 = 3\text{k}\Omega$; נכון $R_3 = 1\text{k}\Omega$; נכון $R_1 = 2\text{k}\Omega$
 $R_T = R_{1,2,3} = \frac{3 \cdot 3}{3 + 3} = \frac{3}{2}\text{k}\Omega$; נכון $R_2 = 3\text{k}\Omega$; נכון $R_{1,3} = 3\text{k}\Omega$

הנני מנסה להחליט על הזרם I_{R1} ו- I_{R2}

$I_{R1,3} = 3 \cdot \frac{3}{3+3} = 1.5\text{A}$; $I_{R1,3}$

$\downarrow$
 $I_{R1} = I_{R3} = 1.5\text{A}$

נכון $R_3 = 1\text{k}\Omega$; נכון $R_1 = 2\text{k}\Omega$

$\downarrow$
 $I_{R2} = 3 \cdot \frac{3}{3+3} = 1.5\text{A}$; I_{R2}

הנני מנסה

I_{S1}	I_{S2}	I_{S3}	I_{R1}
2mA	3mA	3mA	I_{R1}
1mA	1.5A	2mA	I_{R2}
1mA	1.5A	2mA	I_{R2}
3mA	1.5A	5mA	I_{R3}

(3)

$$I - 10 = 20 + 60I_{22} - 24I_{33} - 3I_{22} - 4I_{33}$$

$$I - 30 = 57I_{22} - 28I_{33}$$

$$III \quad 8 = 14I_{33} - 6\frac{2}{3} - 20I_{22} + 8I_{33} - 6I_{22}$$

$$III \quad 14\frac{2}{3} = 22I_{33} - 26I_{22}$$

$$26I_{22} = 22I_{33} - 14\frac{2}{3} \quad | : 26$$

$$I_{22} = \frac{11}{13}I_{33} - \frac{22}{39}$$

: I dkkur 2.3)

$$-30 = 57\left(\frac{11}{13}I_{33} - \frac{22}{39}\right) - 28I_{33}$$

$$-30 = 48\frac{3}{13}I_{33} - 32\frac{2}{13} - 28I_{33}$$

$$2\frac{2}{13} = 20\frac{3}{13}I_{33}$$

$$I_{33} = \frac{28}{263}A$$

: I dkkur 2.3)

$$I_{22} = \frac{11}{13} \cdot \frac{28}{263} - \frac{22}{39}$$

$$I_{22} = -0.474$$

$$I_{22} = 0.474A$$

: I dkkur 2.3)

$$I_{11} = \frac{5}{3} + 5 \cdot 0.474 - 2 \cdot \frac{28}{263}$$

$$I_{11} = -0.916$$

$$I_{11} = 0.916$$

: I dkkur 2.3)

$$I_{R1} = I_{11} = 0.916A \quad : I_{R1}$$

$$I_{R2} = I_{11} - I_{33} = 0.916 - \frac{28}{263} = 1.022A \quad : I_{R2}$$

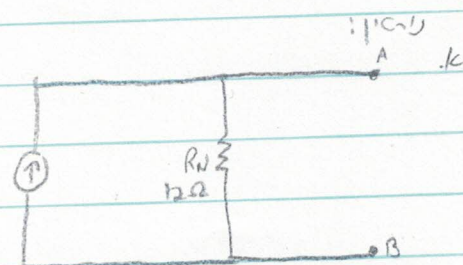
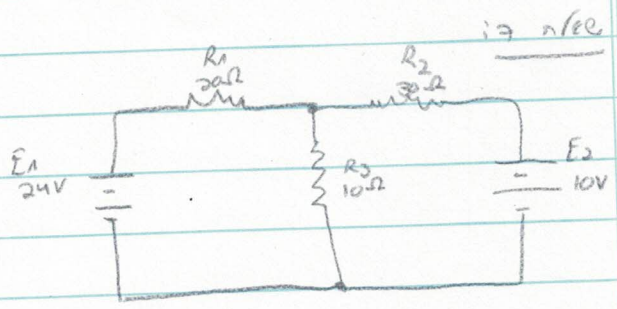
$$I_{R3} = I_{11} - I_{22} = 0.916 - 0.474 = 0.442A \quad : I_{R3}$$

$$I_{R4} = I_{22} = 0.474A \quad : I_{R4}$$

$$I_{R5} = I_{22} - I_{33} = 0.474 - \frac{28}{263} = 0.58A \quad : I_{R5}$$

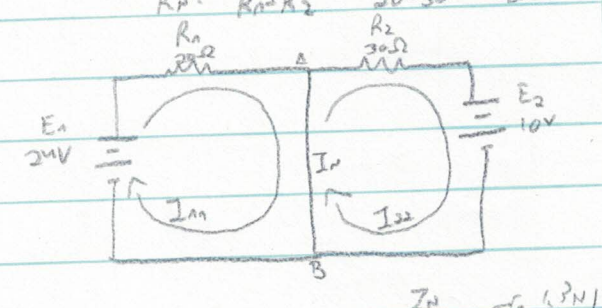
$$I_{R6} = I_{33} = \frac{28}{263}A \quad : I_{R6}$$

③



$$R_N = R_1 \parallel R_2$$

$$R_N = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{20 \cdot 30}{20 + 30} = 12 \Omega$$



$$I \cdot E_1 = I_{N1} (R_1 + R_2) - I_{N2} R_2$$

$$24 = I_{N1} \cdot 20$$

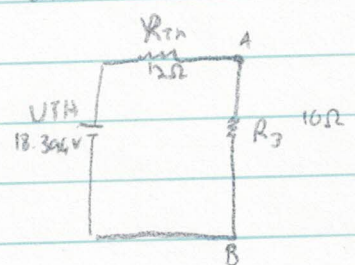
$$I_{N1} = 1.2 A$$

$$II: E_2 = I_{N2} \cdot R_2$$

$$-10 = 30 I_{N2}$$

$$I_{N2} = -\frac{1}{3} A$$

$$I_N = I_{N1} - I_{N2} = 1.2 - (-\frac{1}{3}) = 1.533 A$$



$$R_{TH} = R_N$$

$$V_{TH} = E (V_N)$$

(5)

VAB at (P_N) 11.3V, 10.12 (P_N)

$$I_{R3} = I_1 \cdot \frac{R_1}{R_3 + R_1}$$

$$I_{R3} = 1.553 \cdot \frac{12}{12 + 10}$$

$$I_{R3} = 0.836 A$$

$$U_{R3} = R_3 \cdot I_3$$

$$U_{R3} = 10 \cdot 0.836$$

$$U_{R3} = 8.36 V$$

max (P_N) of U_{TH} (P_N)

$$U_{R3} = U_{TH} \cdot \frac{R_3}{R_3 + R_{TH}}$$

$$8.36 = U_{TH} \cdot \frac{10}{22}$$

$$U_{TH} = 18.392 V$$

1.930A (P_N) (2)

$$P_{R3} = I_{R3}^2 \cdot R_3$$

$$P_{R3} = 0.836^2 \cdot 10$$

$$P_{R3} = 6.98 W$$

$$R_3 = R_{TH} = 11.2 \quad (2)$$

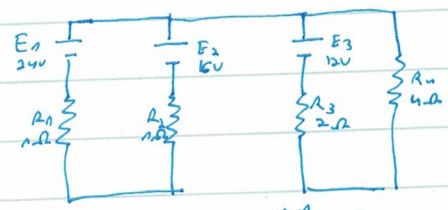
$$U_{TH} = 18.392$$

$$P_{R3} = \frac{18.392^2}{4 \cdot 11.2} = 7.05 W$$

1207

1.8 n/ko

(6)



$$R_{1,2} = \frac{1 \cdot 1}{1+1} = \frac{1}{2} \Omega$$

$$R_{3,4} = \frac{2 \cdot 4}{2+4} = \frac{8}{6} \Omega$$

$$R_{TH} = R_{1,2} + R_{3,4} = 1.833 \Omega$$

$$I_{1,2} = \frac{24-6}{2} = 9A$$

$$E_{1,2} = E_1 - I_{1,2} \cdot R_1 = 24 - 9 \cdot 1 = 15V$$

$$E_{TH} = 15 - 9 = 6V$$

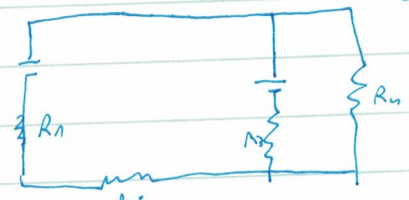
$$I_{R3} = \frac{12}{2+4} = 2A$$

$$I_{3,4} = \frac{12}{2+4} = 2A$$

$$E_{3,4} = 2 \cdot 4 = 8V$$

$$I_{R5} = \frac{12}{1.833+2.08} = 3A$$

(7)



$$R_{3,4} = \frac{1 \cdot 4}{1+4} = \frac{4}{5} A$$

$$R_{3,4} + R_5 = 3.5 \Omega$$

$$R_{TH} = \frac{1 \cdot 3.5}{1+3.5} = 0.7 \Omega$$

$$E_{TH} = \frac{4V}{4}$$

$$I_{R2} = \frac{4V}{0.7+1} = 2.5A$$

$$I_{R5} = \frac{12}{1.833+1.166} = 4A$$

(8)

... $P_{R2} P_{R3} > 6 \text{ n/ko}$

שיטת זרמי הזגים - המלך

• תרגול:

תרגיל 5 שאלה 6

נתון: $I_{n1} = 2A$ חייב להיות $I_{n1} = 2A$ כי זהו הזרם הנכנס לנוד $n1$

$$\begin{aligned} I_{R1} &= I_{n1} = 2A \quad \uparrow \\ I_{R2} &= I_{n1} - I_{R1} = 0.94A \rightarrow \\ I_{R3} &= I_{n1} - I_{R2} = 1.513A \downarrow \\ I_{R4} &= I_{R2} - I_{R3} = 0.439A \leftarrow \\ I_{R5} &= I_{R3} = 1.513A \downarrow \\ I_{R6} &= I_{R3} = 1.513A \downarrow \end{aligned}$$

כדי להימנע מטעות

$$I_{R1} = V \cdot I = 2 \cdot V$$

$$U_{AB} = U_{R1} + U_{R2} + U_{R3} - E_3$$

$$U_{AB} = 0.4 \cdot 2 + 0.4 \cdot 0.94 + 0.4 \cdot 1.513 - 10 = -10.4$$

$$U_{AB} = 466.1V$$

$$P_{R1} = 2 \cdot 466.1 = 932.2W$$

$$P_{R2} = E_3 \cdot I = 100 \cdot 1.513$$

$$P_{E1} + P_{R1} + P_{R2} + P_{R3} + P_{R4} + P_{R5} + P_{R6}$$

$$(I_{R1} - I_{R2} \text{ סוף})$$

Ⓢ

$$I_{22} : -E_3 = I_{22} (R_3 + R_4 + R_5) - I_{n1} R_1 - I_{n2} R_2$$

$$I_{22} = -100 = I_{22} (100 + 50 + 50) - 50 I_{33} - 100 I_{n1}$$

$$I_{33} : -E_4 = I_{33} (R_4 + R_5 + R_6) - I_{n1} R_1 - I_{n2} R_2$$

$$I_{33} = -100 = I_{33} (50 + 200 + 100) - 200 I_{n1} - 50 I_{n2}$$

$$I_{11} : -100 = 30 I_{22} - 50 I_{33} - 200$$

$$I_{11} = 100 = 30 I_{22} - 50 I_{33}$$

$$I_{22} : -100 = I_{22} (R_3 + R_4 + R_5) - I_{n1} R_1 - I_{n2} R_2$$

$$300 = 350 I_{22} - 50 I_{33} \quad / \cdot 10$$

$$1700 = 2100 I_{22} - 300 I_{33}$$

$$1900 = 2050 I_{33}$$

$$I_{33} = 0.926A$$

$$I_{22} = 0.487A \quad (I_{22} \text{ יחידי})$$